

Iguanas and relatives

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Lacertilia
SUPERFAMILY	Iguania
FAMILIES	14
SPECIES	1,840

Iguanas and their relatives include some of the most colorful and familiar of all lizards. Along with geckos and snake-lizards, they are also the most primitive. They are divided into 3 groups: iguanids, agamids, and chameleons. Many species, notably chameleons, are capable of changing their skin color. Some are highly territorial or have elaborate social rituals. Iguanids and agamids look similar to each other but are found in different parts of the world: iguanids occur throughout the Americas, in Madagascar, and on a few Pacific islands, while agamids are found in Africa, Asia, and Australasia. Chameleons occur in Africa, Madagascar, Asia, and Europe. Iguanas and their relatives occupy various habitats, including desert, grassland, and rain forest. Most species lay eggs but a small number give birth to live young.

Anatomy

Iguanids and agamids look alike and occupy similar ecological niches. The main difference between them is in the teeth: unlike iguanids, the agamids have teeth that are fixed to the jawbone and not replaced if lost. Most species have small scales and a loose fold of skin (or dewlap) under the head. The males are often brightly colored, and many species have ornamentation on their head and a crest down their back. They may also have a brightly colored throat fan, which they use for communication.

Chameleons are adapted to life in the trees. Their narrow body and leaflike profile help make them inconspicuous among foliage, while a prehensile tail and toes that are fused in 2 opposing groups enable them to hold tightly to branches. Chameleons also have a well-known (but often exaggerated) ability to change color (see below). Male chameleons usually have a crest or horns on their head.

Color change

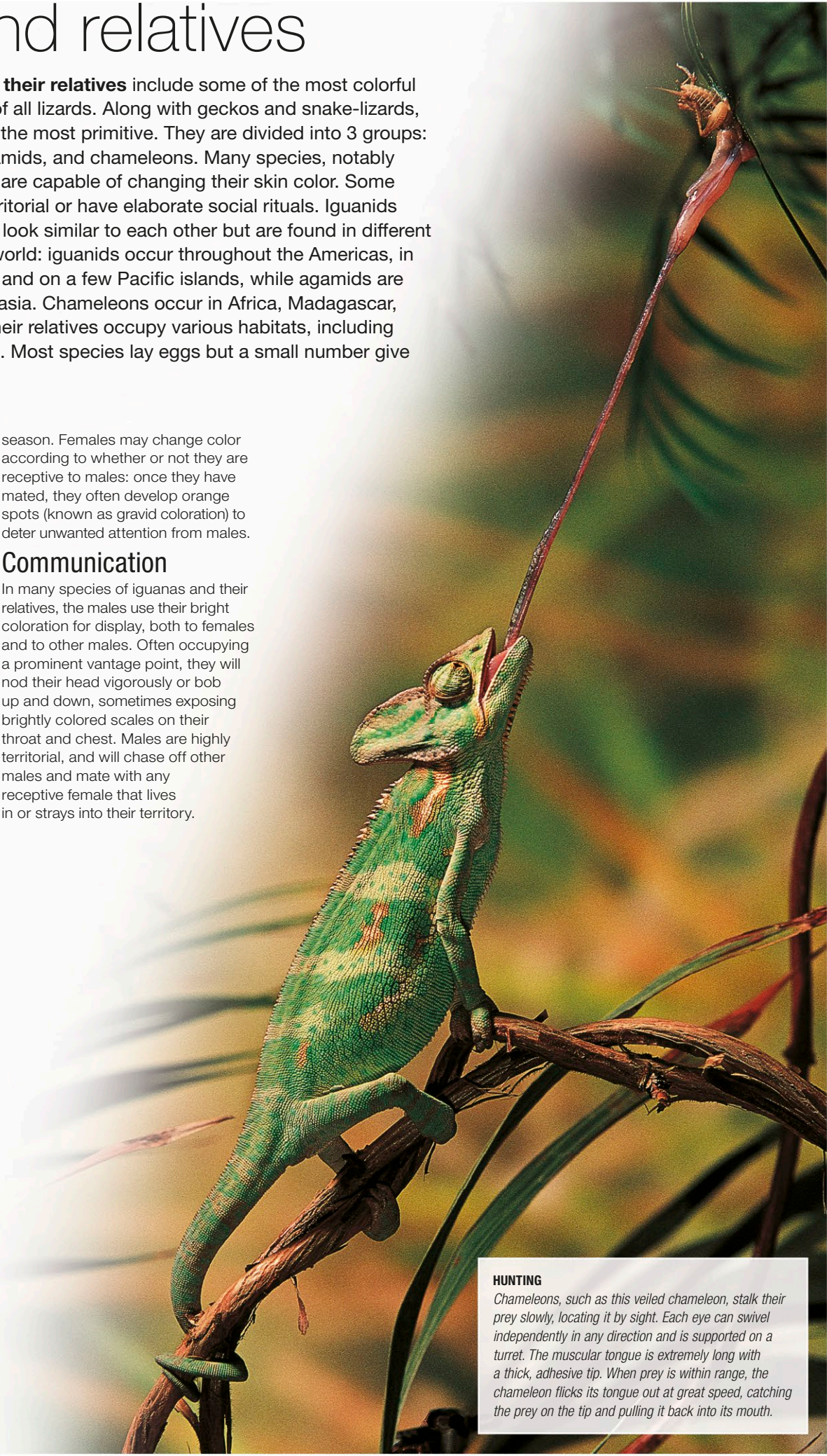
Two types of color change can be seen in the lizards in this group. Chameleons change color by expanding or contracting pigment cells (chromatophores) scattered in their skin. When the cells are expanded, the skin darkens due to the dispersal of melanin pigments in the cells; as the cells are contracted, the skin becomes lighter. A great number of different colors and patterns can be produced in this way. Chameleons change color mainly to control their temperature (making themselves darker when they need to warm up) or to communicate with other members of their species, rather than for camouflage. At night, while they are asleep, chameleons always become paler. Small leaf chameleons and pygmy chameleons, which are brown, have limited ability to change color.

Agamids and iguanids also change color, but this takes place more slowly than in chameleons. Mature males become brighter in the breeding

season. Females may change color according to whether or not they are receptive to males: once they have mated, they often develop orange spots (known as gravid coloration) to deter unwanted attention from males.

Communication

In many species of iguanas and their relatives, the males use their bright coloration for display, both to females and to other males. Often occupying a prominent vantage point, they will nod their head vigorously or bob up and down, sometimes exposing brightly colored scales on their throat and chest. Males are highly territorial, and will chase off other males and mate with any receptive female that lives in or strays into their territory.



HUNTING

Chameleons, such as this veiled chameleon, stalk their prey slowly, locating it by sight. Each eye can swivel independently in any direction and is supported on a turret. The muscular tongue is extremely long with a thick, adhesive tip. When prey is within range, the chameleon flicks its tongue out at great speed, catching the prey on the tip and pulling it back into its mouth.